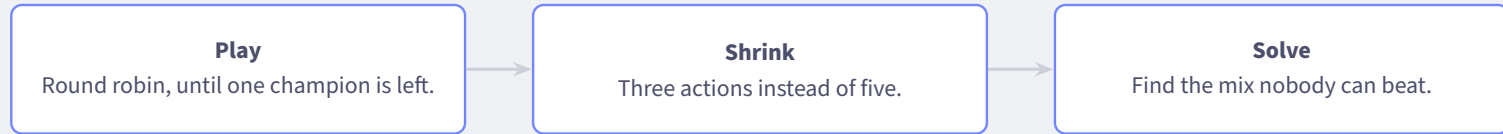


# Nash equilibrium

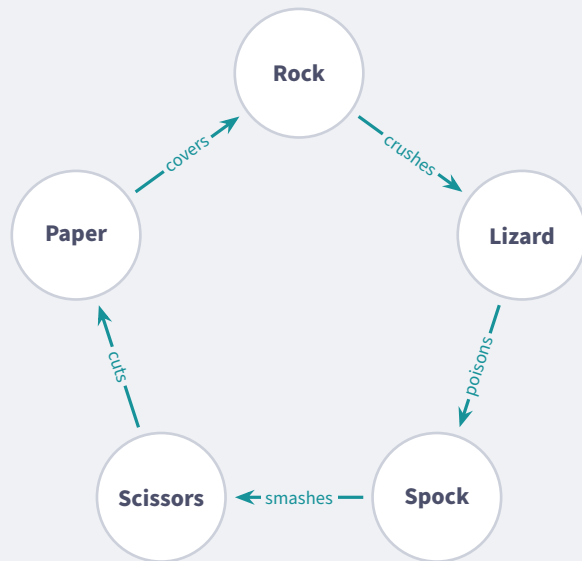
Rock Paper Scissors Lizard Spock, all the way down



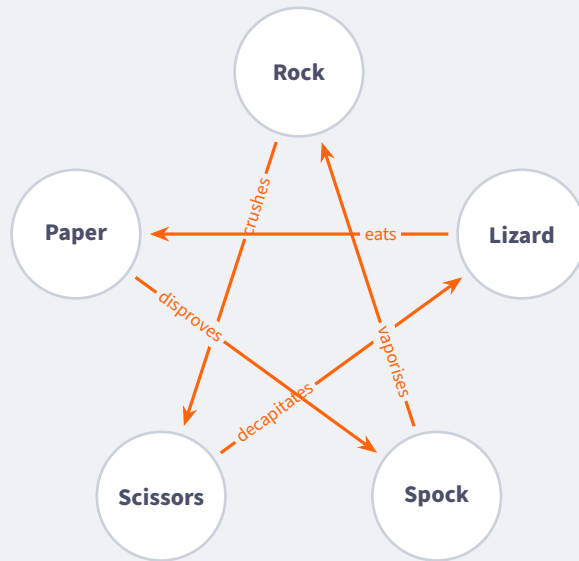
Class champion:

Did I beat them?

### Beat the one beside you



### Beat the one across from you



Each action wins , loses  and draws . So the game is \_\_\_\_\_, and every cell's payoffs add to .

|          | Rock | Paper | Scissors |
|----------|------|-------|----------|
| Rock     |      |       |          |
| Paper    |      |       |          |
| Scissors |      |       |          |

Fill in the *row* player's payoff:

win = +1, draw = 0, lose = -1.

Then circle each player's best responses.

The column player's matrix is .

A cell with both players' best responses circled would be a pure Nash equilibrium.

If they play Rock, I want

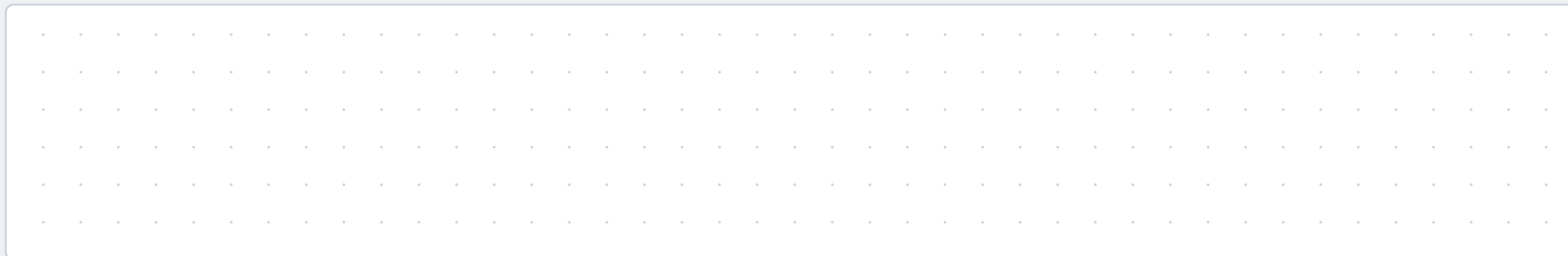
If they play Paper, I want

If they play Scissors, I want

... and then they want to move too.

**So any equilibrium has to be \_\_\_\_\_.**

A mixed strategy  $\sigma$  is a best response to  $\tau$  if and only if



In words: every action I actually use has to be \_\_\_\_\_.

So if I mix over all three actions, all three must give me the same expected payoff.

$$u_r(\text{Rock}, \sigma_c) =$$
$$u_r(\text{Scissors}, \sigma_c) =$$

Support enumeration checks every pair of supports in turn.

Non-empty supports for one player, out of 3 actions:

Pairs of supports to check:

And with 5 actions, as in the game we played:

Which is why we want theorems, not enumeration.

What we just did in the room is **Question 1** on the Nash Equilibrium page.

- (a) Why it is symmetric and zero sum, and the column player's matrix. [3]
- (b) Utilities for three given pairs of mixed strategies. [3]
- (c) Use the best response condition to prove  $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  is an equilibrium. [5]
- (d) Prove it is the *unique* equilibrium. [5]
- (e) State and justify the equilibrium for the five-action game. [3]

[vknights.org/gt/topics/nash-equilibrium.html](http://vknights.org/gt/topics/nash-equilibrium.html)